

Woxsen University

School of Technology

B.Tech. 2nd year (2025-29)

SEMESTER-

Internal Test-I

Course: Data Analytics and Visualization	Course Code: 25TU05MJA3
Program: B-Tech_AIML	Date: 13/08/2026
Max. Marks: 15	Max. Time: 1 hour

Name of the Student:

Roll No.:

Instructions:

1. Write your roll number on the question paper at the place specified for it.
2. Ensure that you have entered all the details in the answer book and have obtained Invigilator's signature.
3. Do not write/mark anything on the question paper.
4. Usage of mobiles and electronic gadgets is strictly prohibited.

Section – A (All questions are Compulsory; Each question carries 1 marks)

(3×1=3)

Use Microsoft Excel for calculations.

Write the Excel formula used wherever applicable.

Assume the dataset is available in Excel Sheet1.

Q.No.	Question	CO	Blooms Taxonomy Level	Marks
Q.1	Using the given Healthcare Dataset , write the Excel formula to calculate the Mean Age .	1	L2 (Understand)	1
Q.2	Using the same dataset, write the Excel formula to replace missing values in the Age column using the Mean.	1	L2 (Understand)	1
Q3	Write the Excel formula to calculate the Correlation between BMI and Blood Sugar.	2	L2 (Understand)	1

Section – B (Answer any 2 questions; Each question carries 2 marks)

(2×2=4)

Q.No.	Question	CO	Blooms Taxonomy Level	Marks
Q.4	Using the Healthcare Dataset, identify all missing values and duplicate records. Remove the duplicate records and replace the missing values in the Gender column using an appropriate method. Submit the cleaned dataset.	1	L3 (Apply)	2
Q.5	Using the Retail Sales Dataset, create a Pivot Table showing the Total Sales by Region and generate a Clustered Column Chart. Identify the region with the highest sales.	1	L3 (Apply)	2
Q 6	Using the Healthcare Dataset, perform Exploratory Data Analysis (EDA) by generating descriptive statistics (using Data Analysis ToolPak or Descriptive Statistics option) for the numerical attributes. Comment on any two observations from the output.	2	L3 (Apply)	2

Section – C (Answer any 2 questions; Each question carries 4 marks)

(2×4=8)

Q.No.	Question	CO	Blooms Taxonomy Level	Marks
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Q.7	Using the Healthcare Dataset, perform a One-Way ANOVA to determine whether there is a significant difference in the Blood Sugar levels among different Diagnosis groups. State the Null Hypothesis (H_0) and Alternative Hypothesis (H_1), generate the ANOVA output in Excel, and interpret the result at a 5% significance level.	2	L4 (Analyze)	4
Q.8	Using the Retail Sales Dataset, calculate the Pearson Correlation between Advertisement Spend and Sales Revenue. Create a Scatter Plot with a Trendline, report the correlation coefficient, and interpret the relationship between the two variables.	2	L4 (Analyze)	4
Q.9	Using the Healthcare Dataset, detect outliers in the Blood Sugar column using the Interquartile Range (IQR) method. Calculate Q1, Q3, IQR, Lower Bound, and Upper Bound, identify the outliers, and create scatter plot.	2	L4 (Analyze)	4
Q.10	Using the Retail Sales Dataset, perform Exploratory Data Analysis (EDA) by generating descriptive statistics for all numerical attributes. Create suitable visualizations (Histogram or Box Plot) for any one numerical attribute and summarize three key business insights from the analysis.	2	L4 (Analyze)	4

-----End of the Question Paper -----